Semiconductor Model Manual v2.0

A Self-Consistent 2D-1D Wave-Optical Laser Simulator Manual

1. Domain & Electrostatic Poisson Equation

The simulation covers the 2D transverse plane (y,z) of a edge-emitting ridge-waveguide diode laser.
Poisson's equation determines the electrostatic potential (psi) and electric field driving carrier transport:

$$-\mathrm{div}(\varepsilon \cdot \mathrm{grad}(\psi)) = q \cdot (p - n + N_D - N_A)$$

Where:

- $\varepsilon = \varepsilon_0 \cdot \varepsilon_r$ is the local permittivity.
- n and p are the electron and hole densities.
- N_D and N_A are the donor and acceptor doping profiles.

2. Carrier Conservation & Recombination Sinks

The carrier conservation equations govern electron and hole flow:

$$\mathrm{div}(J_n) = q \cdot R_{total} \quad \text{and} \quad -\mathrm{div}(J_p) = q \cdot R_{total}$$

where the total recombination sink in the active MQW region is:

$$R_{total} = R_{SRH} + R_{sp} + R_{Auger} + R_{stim}$$

- Shockley-Read-Hall: $R_{SRH} = (n p - n_i^2) / [t_p(n+n_i) + t_n(p+n_i)]$
- Spontaneous Radiative: $R_{sp} = B_{sp} \cdot (n p - n_i^2)$
- Auger Loss: $R_{Auger} = (C_n n + C_p p) \cdot (n p - n_i^2)$
- Stimulated Emission: $R_{stim} = v_g \cdot g(y,z,n,p,T) \cdot s_{ph}(y,z)$

Here, $s_{ph}(y,z)$ is the local photon density from the 2D PhotonSolver.

3. 2D Photon Transport Solver

To account for spatial photon spreading and local conservation, PhotonSolver solves the real-valued diffusion-reaction equation for the photon density $s_{ph}(y,z)$:

$$-\mathrm{div}(D_{ph} \cdot \mathrm{grad}(s_{ph})) + \alpha_{decay} \cdot s_{ph} = R_{source}$$

Where:

- D_{ph} is the photon diffusivity (modeled as 1.0e-3).
- $R_{source} = \beta_{sp} \cdot R_{sp} + R_{stim}$ is the local generation source.
- $\alpha_{decay} = v_g \cdot \alpha_i + v_g \cdot \alpha_{escape}$ is the decay rate.

The escape loss α_{escape} represents the longitudinal facet emissions, tuned dynamically by the 1D Longitudinal Solver based on cavity reflectivities.

4. Complex Transverse Helmholtz Solver (EMSolver)

Using the variable separation $E(x,y,z) = E_{1d}(x) \cdot E_{2d}(y,z)$, EMSolver solves the 2D transverse wave equation:

$$\left[\frac{d^2}{dy^2} + \frac{d^2}{dz^2} \right] E_{2d} + k_0^2 \cdot \tilde{\epsilon} \cdot E_{2d} = \tilde{\beta}^2 \cdot E_{2d}$$

This is formulated as a parameterized eigenvalue problem:

- Eigenvector: The transverse mode profile $E_{2d}(y,z)$.
- Eigenvalue: $\lambda = \tilde{\beta}^2$, where $\tilde{\beta} = \beta_r + i\beta_i$.
- Complex Permittivity: $\tilde{\epsilon} = n_r^2 - i \cdot (n_r \cdot c / \omega) \cdot [\, g(y,z) - \alpha_i - \alpha_{cav} \,]$

The real part n_r acts as a waveguide well, while the spatial mismatch of the imaginary part drives a transverse power flow that shapes the mode phase fronts.

5. Refractive Index Gradients

The real index is a dynamic function: $n_r(y,z) = n_{bulk} + dn_T + dn_C$.

- $dn_r / dT = 2.0e-4 / K$ (positive thermo-optic coefficient causing thermal lensing).
- $dn_r / dn = -2.5e-20\ cm^3$ (negative carrier coefficient causing plasma/band-filling index reduction).

These perturbations continuously update the Helmholtz potential well, driving a self-consistent mode shape update during simulator iterations.

6. Lattice Heat Transport

Lattice temperature is governed by the heat transport equation:

$$-\mathrm{div}(k_T * \mathrm{grad}(T)) = Q_{gen}$$

Where the heat source Q_{gen} combines:

- Joule heating: $Q_{Joule} = J . E0$
- Recombination heating: $Q_{recomb} = E_{tr} * (R_{SRH} + R_{Auger})$
- Optical absorption heating: $Q_{abs} = E_{tr} * v_g * alpha_{abs} * s_{ph}$

Cooling is applied through package boundary convective heat transfer conditions.

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